

170S - Chapters 8 and 9 review list - Spring 2019

Some of the key definitions and ideas that you are expected to know:

- 8.1 **definitions:** null and alternative hypotheses, simple and composite hypotheses, Type I and II error, significance level, test statistic, critical region, p-values
how to: compute all of the above by definition (examples 8.1-1, 8.1-2);
how to test hypothesis about one mean (in the cases of known and unknown variance) in 3 ways: using critical regions, using p-values and using confidence intervals (see pp 361-362 for the discussion);
how to compute critical region of given significance for testing hypothesis for one mean (3 options for 3 types of alternative hypotheses)
- 8.2 **how to test hypothesis about two means (in the cases of known and unknown variance)**
how to compute critical region of given significance for testing hypothesis for two means (3 options for 3 types of alternative hypotheses)
- 8.3 same for the tests about proportions and comparison of two proportions
- 8.4 **how to:** perform sign and Wilcoxon tests, for the median of a distribution and for comparison of two distributions. You should know assumptions for Wilcoxon test (and what to do when two observations happen to equal to each other or to the median)
- 8.5 **definitions:** power function of the test
how to: sketch its plot; find the values at specific points (also, be clear on the connection between power function and type I and II errors)
- 8.6 **definitions:** best critical regions, uniformly best critical regions, (uniformly) most powerful test.
how to: use Neyman-Pearson lemma to find all of the above
- 9.1, 9.2 **how to:** perform chi-square goodness of fit test for one distribution (proportions of data in each of k classes, condition $p_k n > 5$ and why do we need it); for homogeneity and for the independence of 2 attributes
- 9.3, 9.4 **definitions:** grand mean, total sum of squares, error sum of squares, between-treatment sum of squares, mean sums of squares, ANOVA tables; interactions between the two factors influencing experiment
how to: perform ANOVA test for one-factor and two-factor experiments (in the case of one observation per class and no interactions); know mathematical conditions for the no interactions model; observe from data if no interactions model makes sense